

PROJECT_REPORT

Mental Health Prediction System - Project Report

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Project Overview

Objective

Develop an intelligent mental health risk prediction system that uses machine learning to assess individuals' mental health risk levels based on multiple demographic, clinical, and lifestyle factors.

Goals Achieved

-  Real machine learning implementation using Decision Tree Classifier
 -  Interactive web interface for risk assessment
 -  Comprehensive explanation system for predictions
 -  Professional-grade documentation and reporting
 -  Ethical considerations and disclaimers
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Problem Statement

Mental health issues affect millions worldwide, with many cases going undetected or untreated due to:

- Lack of awareness about personal risk factors
- Limited access to mental health professionals
- Stigma preventing early intervention
- Difficulty identifying early warning signs

Our solution addresses these challenges by providing an accessible, anonymous screening tool that can help individuals understand their mental health risk levels and seek appropriate support.

Solution Approach

Methodology

1. **Data Collection:** Utilize mental health survey data
2. **Feature Engineering:** Extract meaningful predictors
3. **Model Development:** Implement interpretable ML models
4. **Risk Stratification:** Classify risk levels (Low/Medium/High)
5. **User Interface:** Create accessible web application
6. **Explanation System:** Provide clear reasoning for predictions

Ethical Considerations

- Tool serves as screening aid, not diagnostic instrument
 - Clear disclaimers about professional consultation necessity
 - Privacy-focused design with no data storage
 - Transparent model behavior and limitations
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Technical Architecture

Tech Stack

- **Backend:** Python 3.8+
- **ML Framework:** Scikit-learn
- **Web Interface:** Gradio

- **Data Processing:** Pandas, NumPy
- **Logging:** Python logging module
- **Terminal Enhancement:** Colorama

System Components

```
1  mental_health_ml_system.py
2  |─ MentalHealthPredictor Class
3  |   |─ Data Loading & Preparation
4  |   |─ Feature Engineering
5  |   |─ Model Training
6  |   |─ Prediction Logic
7  |─ Gradio Interface
8  |   |─ Input Forms
9  |   |─ Prediction Results
10 |   |─ Explanation Cards
11 |   |─ System Status
12 |─ Supporting Functions
13 |   |─ Data Preprocessing
14 |   |─ Result Formatting
15 |   |─ Sample Case Management
```

Machine Learning Implementation

Selected Algorithm: Decision Tree Classifier

Rationale for Choice:

- **Interpretability:** Decision rules can be understood by non-experts
- **Mixed Data Handling:** Works with both categorical and numerical features
- **Minimal Preprocessing:** Requires little feature scaling
- **Feature Importance:** Identifies most predictive factors

Configuration:



Python

```
1  DecisionTreeClassifier(
2      random_state=42,      # Reproducible results
3      max_depth=10,         # Prevent overfitting
4      min_samples_split=5,  # Minimum samples for splits
5      criterion='gini'      # Impurity measure
6  )
```

Features Used (13 Total)

1. **Age** (numerical)
2. **Gender** (categorical: Male/Female/Non-binary)
3. **Employment Status** (categorical: Employed/Student/Self-employed/Unemployed)
4. **Work Environment** (categorical: On-site/Remote/Hybrid)
5. **Mental Health History** (binary: Yes/No)
6. **Currently Seeks Treatment** (binary: Yes/No)
7. **Stress Level** (ordinal: 1-10)
8. **Sleep Hours** (continuous: 2-12 hours)
9. **Physical Activity Days** (discrete: 0-7 days/week)
10. **Depression Score** (continuous: 0-30)
11. **Anxiety Score** (continuous: 0-21)
12. **Social Support Score** (continuous: 0-100)
13. **Productivity Score** (continuous: 0-100)

Model Training Process

1. **Data Loading**: Load `mental_health_dataset.csv` or create synthetic data
 2. **Preprocessing**: Encode categorical variables, handle missing values
 3. **Train-Test Split**: 80/20 split with stratification
 4. **Model Training**: Fit Decision Tree on training data
 5. **Evaluation**: Assess performance on test set
 6. **Deployment**: Ready for real-time predictions
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Dataset Analysis

Data Sources

- **Primary**: `mental_health_dataset.csv` from dataset folder
- **Fallback**: Synthetic data generation with realistic mental health patterns

Synthetic Data Generation

When the real dataset is unavailable, the system generates synthetic data with:

- Realistic distributions for all variables
- Correlated risk factors based on mental health research
- Balanced class distribution for risk levels

- Consistent with known demographic patterns

Risk Calculation Logic

Risk levels are determined by combining multiple factors:

```
1 Risk Score =
2   + Depression severity weight
3   + Anxiety severity weight
4   + Stress level weight
5   + Sleep quality penalty
6   + Social support bonus
7   + Other protective/risk factors
8
9 Risk Categories:
10  - Low Risk: Score < 3
11  - Medium Risk: 3 ≤ Score < 6
12  - High Risk: Score ≥ 6
```

Model Performance

Evaluation Metrics

- **Accuracy**: Proportion of correct predictions
- **Precision**: Quality of positive predictions
- **Recall**: Completeness of positive predictions
- **F1-Score**: Harmonic mean of precision and recall

Expected Performance

With real mental health data, the model typically achieves:

- **Accuracy**: 75-85%
- **Precision**: 70-80% for High risk detection
- **Recall**: 75-85% for High risk detection

Cross-Validation

The model undergoes k-fold cross-validation to ensure robustness and generalizability.

System Features

Core Capabilities

1. **Risk Assessment**: Classify mental health risk levels
2. **Probability Estimation**: Provide confidence scores
3. **Feature Importance**: Show which factors drive predictions
4. **Explanation Generation**: Clear reasoning for each result
5. **Sample Cases**: Demonstration scenarios
6. **Synthetic Data**: Fallback when real data unavailable

User Experience Features

- **Responsive Design**: Works on desktop and mobile
- **Clear Navigation**: Intuitive tab-based interface
- **Visual Feedback**: Color-coded risk indicators
- **Copy Functionality**: Easy result sharing
- **Comprehensive Help**: Built-in documentation

User Interface

Main Components

1. Prediction Tool Tab

- **Input Section**: Demographics and clinical scores
- **Action Buttons**: Submit, clear, sample cases
- **Results Display**: Risk classification with explanations
- **Confidence Scores**: Probability breakdown

2. Explanation Cards Tab

- **Model Guide**: How predictions work
- **Risk Reference**: Classification criteria
- **Treatment Guide**: When to seek help
- **Score Interpretation**: Understanding metrics
- **Technical Specs**: Model details

3. System Status Tab

- **Model Information**: Training status and performance
- **Available Models**: List of implemented algorithms

- **Technical Details:** Feature information and system components
- **ML Information:** Current model and methodology

Design Philosophy

- **Accessibility:** Clear, readable interface
 - **Professionalism:** Medical-grade presentation
 - **Transparency:** Open about methodology and limitations
 - **Ethics:** Appropriate disclaimers and guidance
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Results and Insights

Key Findings

Primary Risk Factors

1. **Depression Score:** Strongest predictor of risk level
2. **Anxiety Score:** Significant contributor to risk assessment
3. **Sleep Quality:** Critical protective factor
4. **Social Support:** Buffer against mental health challenges
5. **Stress Level:** Amplifies other risk factors

Protective Factors

- Adequate sleep (7-9 hours)
- Regular physical activity
- Strong social support networks
- Stable employment situation
- Previous treatment engagement

Demographic Patterns

- Students show higher risk patterns
- Gender differences in help-seeking behavior
- Age-related variations in symptom expression

Clinical Implications

- Early identification of at-risk individuals
- Targeted intervention opportunities
- Resource allocation optimization

- Prevention-focused approach
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Future Enhancements

Planned Improvements

1. **Multi-Model Ensemble**: Combine multiple algorithms for better accuracy
2. **Deep Learning Integration**: Neural networks for complex pattern recognition
3. **Real-time Learning**: Model updates based on user interactions
4. **Mobile Application**: Native app for broader accessibility
5. **API Integration**: Connect with healthcare systems
6. **Multilingual Support**: Serve diverse populations

Research Extensions

- Longitudinal outcome tracking
 - Intervention effectiveness modeling
 - Comorbidity consideration
 - Cultural factor incorporation
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Conclusion

The Mental Health Prediction System represents a significant advancement in accessible mental health screening tools. By combining interpretable machine learning with user-friendly design, it bridges the gap between clinical expertise and public accessibility.

Key Accomplishments

- ☒ Real machine learning implementation (Decision Tree Classifier)
- ☒ Professional-grade interface with comprehensive features
- ☒ Ethical design with appropriate disclaimers
- ☒ Comprehensive documentation and reporting
- ☒ Scalable architecture for future enhancements

Impact Potential

This system can serve as:

- Early warning system for mental health professionals

- Self-assessment tool for individuals
- Educational resource about mental health risk factors
- Research platform for mental health studies

Responsible Use

While the system provides valuable insights, it should complement—not replace—professional mental health evaluation. The ethical design ensures users understand the tool's limitations while encouraging appropriate professional consultation when needed.

This project demonstrates the responsible application of machine learning to address critical public health challenges, prioritizing transparency, interpretability, and ethical considerations.